

From PDK & JDA to Silicon: The UM002B Roadmap

Engineering Perspective — the step-by-step plan from
foundry engagement to qualified silicon

Engineering Perspective

September 2026

1. Introduction — where we stand

1.1 Starting point

The UM002B program has completed its pre-silicon Manufacturing Readiness phase (items 16.1–16.4, 17.1–17.2, 18.1–18.3, plus the Phase 4 validation matrix) with a verdict of **PASS WITH HONEST FAILURES**: coherence audit 15/15 clean, TORR v3 at 8 closed / 9 partial / 7 requires-fab, and the sign-off subset at 0 closed / 5 partial / 2 requires-fab. Everything below builds on that baseline — nothing here re-litigates it.

The modeled design point, for reference:

- **Cell:** 2T1C1M. T_DRAM (thin-oxide) + capacitor forms the volatile DRAM path (~2.8 ns access, refresh-maintained). T_NVM (thick-oxide, ~7 nm modeled) + HfO_x memristor forms the nonvolatile shadow (data as resistance, retained without power).
- **Modes:** NORMAL (all host traffic on the DRAM path; memristor idle, off the critical path), SAVE (power-fail event: charge → resistance, row by row, powered by on-board reserve capacitors), OFF (zero standby energy), RESTORE (power-up: resistance → charge, with ECC).
- **Controller:** backup/restore sequencing, power-fail detection, ECC, test access, and retained metadata live in the on-die array periphery.
- **Characterization basis:** backup energy and timing (~34 μs, ~12.5 μJ) were established on the modeled 68×148 array only and must not be extrapolated to product-scale arrays.

1.2 Open device issues to be closed on this roadmap

Three issues are on record as honest failures. Each is re-stated here with its current numbers, because the roadmap's job is to close them — not to redescribe them optimistically.

1. **Cold-corner RESET misses REQ-006 (≤ 30 ns) at -40 °C.** Real-device (SPICE-class) simulation shows 0.00× margin against the 1.5× target — it fails even at 3.3 V with 8× drive strength. Known recovery paths: program voltage of ≥ 4.06 –4.48 V (outside the 3.3 V domain, implying charge-pump circuitry) or a device-level voltage-acceleration factor of $\alpha \approx 0.40$ eV/V (projected 8.44 ns RESET — projection only, pending measured data).

2. **Shared-bitline disturb fails at 85 °C.** Passes cold, marginal at nominal, fails hot. Modeled safe window 2.0–4.45 V at nominal; the disturb threshold sags to ~4.07 V (assumed analytical value) at 85 °C, collapsing the window. All figures are modeled under assumed parameters — not silicon proof.
3. **Far-row IR drop 80–98 mV vs 75 mV budget** at nominal/hot. SET remains functional, but margin is consumed. Buried wordline does not rescue the 132 nm pitch; the standing recommendation is 132×288 nm (47.2 → 48.5 F²). Bounding design rules carried forward: HS-5, IR-5, HS-1 amended 6.4 → 6.8 μs, 20 ns thermal pulse ceiling.

1.3 What the PDK changes, and what the JDA changes

The PDK is descriptive. It supplies silicon-calibrated transistor models, real design rules, the real metal stack, and the foundry's sign-off decks. With it, every simulation, layout, and margin in the program is re-baselined against fab truth. It fixes the *design-closable* portion of the three issues: driver sizing, biasing schemes, wire sizing, pitch, grid.

The JDA is prescriptive. The HfO_x stack's switching kinetics, voltage acceleration, disturb threshold, and retention are properties of materials and factory recipes — they cannot be changed in a layout editor. The JDA buys wafer splits, measurement, model extraction, and iteration with the foundry until the device meets its targets. It fixes the *material* portion. For 100% of the custom-device vision, there is no substitute; the alternative (a node with qualified embedded ReRAM, PDK-only) is a different program and out of scope here.

The two tracks run in parallel wherever possible: Phase 1 (PDK hardening) and Phase 2 (JDA device development) overlap, and both feed Phase 3 (full-chip build).

2. Phase 1 — PDK hardening

Objective: re-baseline the entire design against fab truth; close everything closable in design; quantify the remainder and hand it to the JDA track with numbers, not adjectives.

2.1 PDK install and flow setup

- Acquire the PDK under foundry NDA. Pin the PDK version and place it under version control; document NDA and export-control handling — the PDK is the foundry's confidential property and the program's legal footing starts here.
- Assemble the design flow from PDK-delivered decks: SPICE-class simulator with calibrated compact models, schematic/layout capture, parasitic extraction (PEX) deck, and DRC/LVS decks. Prefer no-cost tooling wherever the PDK supports it; the foundry's sign-off decks are mandatory regardless of tool choice.
- **Regression first:** re-run the existing netlists unmodified against the calibrated models before changing anything. This isolates model deltas from design deltas — if a result moves, you know whether the model or your edit moved it.

2.2 Cell and array re-layout to real rules

- Port the 2T1C1M cell to the real DRC rule set. Select the T_NVM thick-oxide device from the PDK's IO device offerings (verify its voltage tolerance against the program-voltage range under evaluation, including the 4.06–4.48 V recovery option).
- Validate the 132×288 nm pitch recommendation against real rules — confirm or revise with written justification. The buried-wordline experiment's negative result stands; do not re-spend effort there without new evidence.
- Deliverable: a DRC/LVS-clean array tile, with a rule-waiver log (every waiver justified in writing, none silent).

2.3 PEX-based parasitic re-extraction

- Extract with the real metal stack: bitline RC, inter-wire coupling capacitance (the disturb analysis input), and wire resistance (the IR-drop input).
- Back-annotate into the simulation benches and archive the extraction corners alongside the schematics. From this point, no bench runs on assumed parasitics.

2.4 Corner re-simulation

- Full matrix: $-40 / 25 / 85\text{ }^{\circ}\text{C} \times \text{SS} / \text{TT} / \text{FF}$, plus Monte Carlo on the sense and ECC paths for mismatch.
- Re-run the three issue benches with calibrated transistor models:
- RESET timing at $-40\text{ }^{\circ}\text{C}$ against REQ-006 ($\leq 30\text{ ns}$, $1.5\times$ margin target);

- disturb window at 85 °C against the (still assumed, pending JDA measurement) threshold;
- IR drop at nominal/hot on the extracted grid against the 75 mV budget.
- Output: a quantified gap-to-target per issue, in numbers. "Marginal" is retired as a status — every issue gets a value and a margin.

2.5 Per-issue design actions

- **Cold RESET:** upsize the T_NVM driver for cold-corner drive strength using calibrated models. In parallel, evaluate the charge-pump option for the 4.06–4.48 V domain — cost it honestly (level shifters, sequencing, area, standby leakage) before committing. Device- α improvement stays on the JDA track; do not burn design margin chasing a device-physics problem with transistors.
- **Disturb:** evaluate biasing-scheme variants and bitline segmentation using PDK-available devices. Requirement: worst-case unselected-cell bias stays below the disturb threshold with explicit margin at 85 °C. If no biasing scheme closes it with the assumed threshold, that is a *finding*, not a failure — it sets the JDA's disturb-threshold target (§3.1).
- **IR drop:** power-grid strap widening, grid planning, and pitch finalization against the real sheet resistance. Close the 75 mV budget on the extracted grid. Carry forward HS-5, IR-5, HS-1 at 6.8 μ s, and the 20 ns thermal pulse ceiling; re-validate each against the real stack.

2.6 Entry / exit criteria

- **Entry:** PDK installed, version-pinned, NDA documented; prior netlists under version control; per-issue success metrics written down before any editing begins.
- **Exit:** DRC/LVS-clean array tile; corner simulations complete with calibrated models; each open issue re-baselined with a number and explicitly partitioned into *design-closable* vs *JDA-required*. Nothing proceeds to Phase 3 on assumed models.

3. Phase 2 — JDA device development

Objective: make the HfO_x device real — measured on wafers, modeled from data, frozen as a characterized device in the private flow.

3.1 Device target agreement

Before any wafer runs, agree the targets with the foundry in writing, in numbers:

- **Switching kinetics:** $\alpha \approx 0.40$ eV/V (projected 8.44 ns RESET at target drive — labeled as projection until measured).
- **Disturb threshold at 85 °C:** must clear the worst-case unselected bias selected in §2.5 with explicit margin. The numeric value is set by the Phase 1 biasing choice — the two phases handshake here.
- **Retention:** 10-year at 85 °C as the *design target*; the bake methodology (acceleration model, sample sizes, read criteria) is agreed now, and the proof is collected in Phase 4. Target defined today, evidence later — never conflated.

3.2 Test-chip planning

- This is a device test chip, not the product: **1T1R test arrays** (statistical switching distributions across hundreds of devices — no hero-device reporting), **Kelvin / 4-point structures** (isolate filament resistance from series parasitics), and process-control monitors on every split.
- **Stack splits:** HfO_x thickness, electrode material/workfunction, oxygen stoichiometry, post-deposition anneal conditions. Each split is a hypothesis about α , disturb immunity, or retention — logged as such.
- Tester requirements: temperature chuck covering –40 °C to ≥125 °C, fast pulsing (sub-30 ns resolution) for RESET kinetics, and low-current sense for disturb detection.
- Secure the JDA shuttle slot and plan sample counts for statistics, not anecdotes.

3.3 Measurement and Verilog-A extraction

- Measure switching kinetics vs voltage and temperature → extract α and thermal activation energy. Measure disturb threshold vs temperature. Run the agreed retention bake.
- Fit the **Verilog-A compact model** to the measured data. Then QA the model against the data across the full matrix — typical *and* corners. A model that fits only typical is a liability; reject it.
- Deliverable: extracted model + extraction report with residuals, so any engineer can see where the model is trustworthy and where it is thin.

3.4 Iteration spins

- Plan for multiple spins; a single-spin success is the exception in device R&D, not the plan.
- Define per-spin go/no-go against the §3.1 targets *before* the wafers run. Record negative results with the same care as positive ones — a split that fails cleanly teaches the next split.

3.5 Stack freeze

- Freeze the stack recipe, freeze the Verilog-A model, freeze the ReRAM-module design rules. From here the device is treated like a PDK device: characterized, versioned, and trusted within its qualified envelope.
- Hand back to design: re-run the Phase 1 benches with the frozen model and confirm the closure numbers. The loop between Phase 1 and Phase 2 closes here.

3.6 Entry / exit criteria

- **Entry:** JDA signed; §3.1 targets agreed in writing; test-chip shuttle slot secured; tester capability confirmed.
- **Exit:** stack frozen; model extracted and QA'd; all three device targets demonstrated on measured silicon — or a documented waiver with a recovery plan. No waiver is silent.

4. Phase 3 — Full-chip build

Objective: integrate the hardened array tile, the verified controller, and the frozen device model into a signed-off, tapeout-ready chip.

4.1 Integration

- **Array:** tile → full array with row/column decoding, sense amplifiers, and write drivers sized per the §2.5 outcomes.
- **Controller RTL:** power-fail detection, SAVE/RESTORE sequencing, ECC encode/decode, test access ports, and retained-metadata register file. The RTL must be functionally verified (constrained-random + directed tests, including power-fail injection at arbitrary points in a SAVE sequence) before it touches layout.

- **IO ring:** pad ring with ESD protection per the foundry's ESD rules; signal integrity on high-speed interfaces validated against the PDK's IO models.
- **Top-level power grid:** designed to the IR-drop budget closed in §2.5, verified on the extracted full-chip grid — not the tile grid.
- **Clock and reset distribution:** skew and integrity checked; the power-fail detector's own supply domain reviewed (it must stay alive long enough to launch SAVE — its hold-up is a design requirement, not an assumption).

4.2 Physical implementation

- Full-chip DRC/LVS with the foundry decks; zero violations or a written waiver log.
- Full-chip PEX. Re-check: bitline RC at array scale, grid resistance at chip scale, coupling on long routes.

4.3 Sign-off

- **STA** across all corners ($-40/25/85\text{ }^{\circ}\text{C} \times \text{SS/TT/FF}$), including the SAVE/RESTORE sequencing paths — the emergency path gets the same timing rigor as the functional path.
- **Power integrity:** IR-drop and electromigration sign-off with the foundry's decks on the extracted grid.
- **Reliability checks** per foundry requirements (electromigration and other required mechanisms — use the foundry's checklist, not a generic one).
- **Issue re-verification at full-chip level:** all three open issues re-run with extracted full-chip parasitics *and* the frozen JDA device model. An issue is closed only when it passes here — tile-level closure is necessary but not sufficient.

4.4 DFT and test modes

- Array BIST for manufacturing test; scan/ATPG per foundry requirements.
- **SAVE/RESTORE test modes:** forced power-fail injection with snapshot verify, so the tester — and later the qualification lab — can exercise the backup path deterministically rather than hoping a real power cut arrives on schedule.

4.5 Tapeout

- First tapeout goes on an **MPW (multi-project wafer) shuttle:** mask costs shared across projects, which keeps first silicon affordable. Accept the trade — fewer samples, shuttle-schedule timing — with eyes open.

- The bring-up board and the tester program are ready *before* silicon returns. Silicon waiting on a test board is schedule burned for nothing.

4.6 Entry / exit criteria

- **Entry:** frozen device model in the flow; DRC/LVS-clean tiles; controller RTL functionally verified including power-fail injection; written test plan.
 - **Exit:** signed-off GDSII taped out on MPW; bring-up hardware and tester program ready.
-

5. Phase 4 — Silicon validation

Objective: replace every remaining "modeled" and "target" in the evidence ledger with measured data — or write down exactly where reality diverged and what happens next.

5.1 Bring-up

- Power-on, controller communication, then the basic SAVE → OFF → RESTORE loopback on the tester using the §4.4 test modes.
- Shmoo the program voltages: find the real operating window on silicon and compare it against the modeled 2.0–4.45 V window. Record the delta — it calibrates every model downstream.

5.2 Issue-mapped characterization

Each open issue gets a dedicated measurement campaign, mapped one-to-one:

- **Cold RESET:** temperature chamber at $-40\text{ }^{\circ}\text{C}$; measure the RESET time distribution across many cells and dies. Pass criterion: REQ-006 ($\leq 30\text{ ns}$) with the $1.5\times$ margin target, on statistics — not on the best cell.
- **Disturb:** $85\text{ }^{\circ}\text{C}$ hammer testing — repeatedly write selected cells while monitoring unselected neighbors for resistance drift. Compare against the threshold agreed in §3.1.
- **Retention:** accelerated high-temperature bake with Arrhenius extrapolation toward the 10-year/ $85\text{ }^{\circ}\text{C}$ design target, using the methodology agreed with the foundry in §3.1. This is industry-standard inference, and it is labeled as inference — the honest statement is "bake data extrapolates to X," never "ten years measured."

5.3 Debug and re-spin criteria

Decide the rules *before* silicon arrives, while heads are cool:

- **Functional failure** (controller bug, sequencing error, dead test mode) → root-cause, then ECO/metal-layer fix if possible, else full re-spin.
- **Parametric miss** (RESET too slow, disturb too high, drop too large) → classify as *design-fixable* (re-spin with a design change from the Phase 1 playbook) or *device-limited* (back to the JDA track for another stack iteration). The classification, not the disappointment, drives the next step.

5.4 Qualification and the honest claim ladder

- Run the foundry's required **JEDEC-style qualification flow** (high-temperature operating life and related stresses, per the qualification checklist — described here generically because the exact flow is the foundry's to specify). Qualification is what stands between "characterized" and "production-worthy."
 - **The claim ladder** — the discipline this whole program has followed:
 - After Phase 4 characterization: *internal* claims only ("RESET measured at X ns at −40 °C across N cells/dies").
 - After qualification: *product* claims (temperature rating, retention) may be stated externally.
 - No claim moves up a rung without the data behind it. The ledger from the pre-silicon program (proven / modeled / target / requires-fab) stays live through silicon — rows get promoted on evidence, never on optimism.
-

6. Parallel software track

The hardware program above does not block the software program, and the software program must not become an excuse to delay hardware decisions.

- **OS persistent-memory enablement** and the **shutdown/resume firmware** (graceful-shutdown orchestration, clean-shutdown flag, hibernation-style resume) are developed in parallel on emulators and FPGA models of the array — no silicon required.
- **Freeze the host interface spec early** (DDR-style / NVDIMM-N-like command and timing interface). The interface is the contract between the two tracks; once frozen,

hardware and software proceed independently and integrate at defined checkpoints.

- Deliverable of this track: a software stack that is ready to exercise real silicon the week it arrives — not a stack that starts when silicon arrives.

7. Residual risks register

Short, honest, and kept live — a risk not written down is a risk not managed.

1. **JDA partner and schedule.** A JDA is negotiated, not purchased; foundries choose their partners and set their pace. Mitigation: begin engagement early, keep the PDK-only fallback (§1.3) as a live option.
2. **Device target achievability.** $\alpha \approx 0.40$ eV/V, 85 °C disturb immunity, and the retention target are goals, not entitlements. Mitigation: multi-spin budget in the plan; per-spin go/no-go (§3.4).
3. **Retention extrapolation.** Accelerated bake is inference, however standard. Mitigation: label it as such; widen sample sizes; never let marketing outrun the bake data.
4. **First-silicon bugs.** Expect a re-spin; the industry norm, not a failure. Mitigation: MPW keeps the cost survivable; §5.3 rules decided in advance.
5. **Funding phase-gates.** JDA fees, MPW runs, and qualification each need capital. Mitigation: align financing milestones with the phase exits defined above — each exit is a fundable proof point.
6. **Model risk.** Every simulation is only as good as its models. Mitigation: the measured-data checkpoints (§3.3 extraction QA, §5.1 shmoo, §5.2 characterization) are the backstops; no model survives contact with contradicting silicon.

End of document. This roadmap describes the forward plan from PDK acquisition and JDA signature to qualified silicon. All pre-silicon figures cited are the modeled design point; all targets are labeled as targets until measured.